

Acoustic manifestation of Russian word-final devoicing in utterance-medial position

Abstract

The present study examines the acoustic realizations of Russian word-final devoicing in utterance-medial positions, which has not been extensively examined in the previous phonetic literature. Noun words were elicited using a nominal inflection task in which speakers were asked to inflect pseudo-nouns in response to auditory stimuli. The results showed that there were small but significant durational differences between words with underlyingly voiced obstruents and those with underlying voiceless obstruents: the vowel duration before an underlyingly voiced stop was longer than that of its voiceless counterpart. The duration of vocal fold vibration was longer during underlyingly voiced fricatives than during voiceless fricatives. These results suggest that, in accordance with previous studies, voicing neutralization in word-final position in Russian is incomplete. Our new finding is that incomplete neutralization occurs even in utterance-medial positions. Further, a comparison of the current results with previous studies suggests that phonetic details of how incomplete neutralization occurs can be different between phrase-final and utterance-final positions.

1. Introduction

The present experiment examines how voicing contrast is implemented in word-final position in Russian. Russian has a two-way voicing contrast in obstruents (stops and fricatives). Previous phonological studies have claimed that Russian voiced obstruents are devoiced in word-final positions, resulting in complete phonological neutralization (e.g., Kenstowicz 1994, Odden 2005). However, this neutralization may be phonetically incomplete: small but consistent acoustic differences, which are demonstrably perceptible, have been observed between supposedly neutralized sounds in Russian (among others, Dmitrieva *et al.* 2010,

Kharlamov 2012, Kulikov 2012) as well as in other languages (Port and O'Dell 1985, Roettger et al. 2011, 2014 for German; Dinnsen and Charles-Luce 1984 for Catalan; Warner et al. 2004 for Dutch). For Russian, while previous acoustic and perceptual studies documented the incomplete neutralization of word-final devoicing in utterance-final positions (e.g., Kharlamov 2012), less well examined is word-final devoicing in utterance-medial positions. Thus, the aim of the present paper is to fill this gap by documenting word-final devoicing in *utterance-medial* positions, which will advance our understanding of the nature of word-final devoicing and incomplete neutralization.

In the following sections, we will first review the previous studies of incomplete neutralization in general (1.1) and of Russian word-final devoicing in particular (1.2). Section 2 presents an acoustic analysis of word-final devoicing in utterance-medial positions. Section 3 discusses the results of the acoustic analysis. Finally, section 4 concludes the paper.

1.1 Word-final devoicing and incomplete neutralization

In many languages, such as German, Dutch, and Catalan, voiced obstruents devoice in domain-final position (See Myers 2011 for an extensive crosslinguistic review). In languages with final devoicing, it has been traditionally formalized, especially in the generative phonology tradition, that the distinction between voiced and voiceless obstruents is categorically lost in word-final position, and therefore, minimal-pairs that underlyingly differ in a voiceless and devoiced obstruent, such as /luk/ and /lug/, result in complete homophones (e.g., Akishina and Baranovskaia 2010, Kenstowicz 1994, Odden 2005, Wade 2011).

However, a number of experimental studies have found that final devoicing may not be categorical in nature. A classic example is found in German final devoicing (Port and O'Dell 1985). Port and O'Dell (1985) examined German final devoicing using 10 minimal pairs of words produced in isolation (only for stop consonants). Their results showed that there were

significant durational differences between the underlyingly voiced stops and the underlyingly voiceless stops. For example, the vowels preceding the underlyingly voiced stop (e.g., Rad [ʁa:t] ‘wheel’) were on average 15 ms longer than those preceding the underlyingly voiceless stop (e.g., Rat [ʁa:t] ‘council’). Further, those differences are perceptible to the native listeners (although the accuracy being only slightly above the chance-level). Based on those findings, Port and O’Dell argued that neutralization was incomplete. Similar patterns of incomplete neutralization have been identified in many languages, in which small but consistent phonetic differences have been observed between segments that are claimed to be phonologically neutralized (among others, Charles-Luce 1985, Piroth and Janker 2004, Port and O’Dell 1985, Port and Crawford 1989, Roettger et al. 2011 for German final devoicing; Dinnsen and Charles-Luce 1984 for Catalan final devoicing; Ernestus and Baayen 2006, 2007, Warner et al. 2004 for Dutch final devoicing; Dmitrieva 2005, Dmitrieva et al. 2010, Kharlamov 2012, Kulikov 2012 for Russian final devoicing; Braver 2013, Herd et al. 2010 for American English flapping; Fourakis and Port 1986 for American English stop epenthesis, and more).

While the existence of incomplete neutralization seems well established in the literature, what incomplete neutralization means to phonological theory is still in question (See Braver 2013 for a recent summary of theoretical treatments of incomplete neutralization). Also, a body of experiments has revealed that multiple factors can affect phonetic realizations of segments, which can result in spurious cases of incomplete neutralization. Specifically, phonetic realization of purportedly neutralized segments is shown to be task-dependent, including the presence/absence of orthographic input during the experiment (Fourakis and Iverson 1984, Warner et al. 2004, 2006, Kharlamov 2012). Fourakis and Iverson (1984) examined the influence of orthographic input on the acoustic realization of final obstruent devoicing in German. Their results showed that, in the condition with orthographic input (i.e.,

a traditional reading task), significant acoustic differences were observed between supposedly neutralizing obstruents. In contrast, no significant differences were found in the task where the speakers were asked to conjugate verbs without orthographic input (a verbal conjugation task). On the basis of their results, Fourakis and Iverson (1984) concluded that the phonetic differences between the underlying voiced and voiceless obstruents are merely due to the effect of German orthography, which reflects the underlying contrasts in voicing. However, recent studies suggest that the phonetic differences in a neutralizing position cannot exclusively be attributed to the orthographic influence. In their study on German final devoicing, Roettger et al. (2011, see also, 2014) adopted an experimental task, in which nonce words were elicited by using auditory stimuli, eliminating the effects of orthography. The results showed that significant differences were observed in terms of preceding vowel duration and burst intensity. They concluded that, contrary to Fourakis and Iverson's (1984) claim, the phonetic differences between purportedly neutralizing segments are observed even in the conditions without orthographic input. The orthographic effects on final devoicing have been examined in other studies on incomplete neutralization (Warner et al. 2004, 2006, Ernestus and Baayen 2006, Kharlamov 2012), and these studies showed that, in general, acoustic differences are greater in reading tasks than in tasks without orthographic input, but that incomplete neutralization is observed even without the presence of orthographic input.

Beside the effect of orthography, word-specific factors such as word token frequency, (Pierrehumbert 2002, VanDam and Port 2005) and lexical neighborhood density (Wright 2004, Baese-Berk and Goldrick 2009) are shown to affect acoustic realizations, potentially resulting in apparent cases of incomplete neutralization. For example, VanDam and Port (2005) showed that the Voice Onset Time (VOT) of English /t/ is shorter for words with high token frequency than those with low frequency when the native speakers read them in phrasal contexts. Further, Baese-Berk and Goldrick (2009) found that the presence/absence of

minimal pairs among lexical items affects VOT in English. They showed that the VOT of a word-initial /p/ that has a minimally contrasted neighbor (e.g., *pox* is minimally contrasted with *box*) was significantly longer than without such a neighbor (e.g., *posh* vs. **bosh*). These effects may create an illusion of incomplete neutralization, for example, the burst of [t] of the underlying /t/ could be longer than that of /d/, just because the word containing /t/ happens to be less frequent or have a minimal pair.

In short, the studies discussed above suggest that multiple factors may affect the realization of neutralizing segments, and hence should be controlled for in studies of incomplete neutralization. To summarize the question: does incomplete neutralization still hold even (i) without orthographic inputs and (ii) without word-specific phonetic effects? The next section is devoted to the discussion of Russian word-final devoicing that serves as direct background for our study.

1.2 Background on Russian word-final devoicing

Russian is one of the many languages in which the distinction between underlying voiced and voiceless obstruents is neutralized to voiceless in the word-final position (e.g. Kenstowicz 1994, Odden 2005, Akishina and Baranovskaia 2010, Wade 2011). Therefore, the words containing underlyingly voiced or voiceless obstruents in the word-final position are, impressionistically speaking, homophonous (e.g., /luk/ [luk] ‘onion_{NOM.SG.}’ vs. /lug/ [luk] ‘meadow_{NOM.SG.}’).

However, previous studies have observed small acoustic differences between these “homophones” in the neutralizing position (Chen 1970, Pye 1986, Dmitrieva et al. 2010, Kharlamov 2012, Kulikov 2012, Shrager 2012). Using a word-reading task, Dmitrieva et al. (2010) showed that there are significant acoustic differences between underlying voiced and voiceless obstruents in word-final position when the speakers read the words in isolation. For

monolingual speakers of Russian, closure/frication duration of underlying voiceless obstruents is significantly longer than that of underlying voiced obstruents. They also found that the duration of the release portion of the underlying voiceless stop is significantly longer than that of the underlying voiced stops. Shrager (2012) examined dental stops in utterance-final position, and reported similar results to Dmitrieva et al. (2010). Chen (1970) and Kulikov (2012) reported that vowels preceding underlyingly voiced stops are longer than those preceding underlyingly voiceless stops. Kulikov (2012) also found that the difference is present in the first formant (F1) at the edge of the preceding vowel (for stops) in such a way that F1 is lower next to underlyingly voiced stops. The results of these studies suggest that neutralization is incomplete in Russian final devoicing. However, the causes of the incomplete neutralization still remain unclear, because several factors that affect phonetic details (as discussed in section 1.1) were not controlled for in those studies. Since reading tasks involve Russian orthography, the obtained phonetic differences may have been brought about by orthography. Other studies have not controlled factors such as the lexical word frequencies of the stimuli, which can lead to a spurious incomplete neutralization.

In his large scale study of methodological influence on final devoicing in Russian, Kharlamov (2012, 2014) found significant differences in closure/frication duration and the duration of the release portion between underlying voiceless and voiced obstruents by means of a reading task and a picture-naming/fill-in-blank task, in which speakers were asked to produce existing Russian words in isolation. Crucially, since the latter task involved no orthographic input, the observed differences can not readily be attributed to orthographic effects. In addition, it was shown that the speakers produced larger acoustic differences when they were exposed to both members of lexical minimal pairs. However, regardless of whether or not the speakers were exposed to both members of lexical minimal pairs in the stimuli, neutralization was incomplete. In summary, Kharlamov (2012, 2014) found that the observed

acoustic difference cannot readily be accounted for solely by methodological factors. Further, since Kharlamov (2012) controlled for the lexical word frequencies of the target words, the results do not seem to come from the intrinsic differences of the stimuli used, either¹.

To summarize, previous instrumental studies suggest that Russian word-final devoicing is incomplete, and that acoustic differences found between supposedly neutralizing obstruents cannot merely be regarded as the consequence of orthography, the presence or absence of a minimal pair in the lexicon, or lexical token frequency. However, there are still some gaps to be filled.

One of the remaining questions is the acoustic manifestation of word-final devoicing in *utterance-medial* positions. While previous studies extensively documented word-final devoicing in utterance-final positions, less well studied is how the devoicing is implemented in word-final (but utterance-medial) positions. The only study we are aware of is Kulikov (2012), who examined word-final devoicing in utterance-medial position by embedding the target words in a frame sentence. As discussed above, his results showed that, for stops, the underlying voicing is distinguished by the preceding vowel duration and the first formant frequency at the edge of the vowel when the speakers performed a reading task. However, since Kulikov (2012) used a reading task, the results may be due to orthographic influences. Also, since the focus of his study was stop consonants, no systematic analysis has been undertaken on fricative devoicing in utterance-medial position.

Thus, the aim of the present study is to provide a comprehensive acoustic-phonetic description of Russian word-final devoicing in utterance-medial position both for stops and

¹ While the focus of the present study is acoustic aspects of word-final devoicing, there are a few studies focusing on its perception and aerodynamics (Kanibolotskaia 2009, Matsui 2011, Kharlamov 2012). Kanibolotskaia (2009) found that the underlyingly voiced and voiceless bilabial stops (/p, b/) are different in terms of air pressure in the oral cavity. Matsui (2011) demonstrated that “incompletely” neutralized velar stops were perceptually distinguishable by the native listeners of Russian. Kharlamov (2012) extended Matsui’s (2011) observation in a large data set.

fricatives. By adopting the similar strategy proposed in Roettger and his colleagues' studies (2011, 2014) for German final devoicing, we examine both stop and fricative devoicing in utterance-medial word-final position, while doing away with orthographic and lexical influences.

2. Methods

2.1 Participants

Twelve native speakers of Russian, aged 19-22 (Mean age: 20.8, SD: 0.9), participated in the production experiment. Seven were female and five were male. All of the participants were born and raised around Orenburg city in Russia, and none of them reported that they use other languages on a daily basis. Two of the participants reported that their national background is both Russian and Kazakh, although they reiterated that they could only speak Russian.

2.2 Speech materials

All non-palatalized obstruents that have voicing contrast in Russian were examined: bilabial (/p/, /b/), dental (/t/, /d/), and velar (/k/, /g/) stops and labio-dental (/f/, /v/), alveolar (/s/, /z/) and post-alveolar (/ʃ/, /ʒ/) fricatives. The vowels following the target obstruents were /i/, /e/, /a/, /o/ or /u/. 30 minimal pairs (6 consonants × 5 vowels) were used as the stimuli. These were all pseudo-nouns. The current study used nonce words in order to avoid the word-specific phonetic effects reviewed in section 1. All of them were monosyllabic words, consisting of either CVC or CCVC structures. The list of all the target words appears in the appendix. In addition to the target words, 60 fillers were added.

In order to elicit the target words from the speakers, auditory stimuli were first produced by a native speaker of Russian (a female in her twenties, from Orenburg). The details of the auditory stimuli are described more in section 2.8.

2.4 Procedures

The general elicitation method followed that of Roettger *et al.* (2011, 2014). Within each recording, one auditory stimulus was first presented to the participants via the headphones (SONY MDR-Z700). The auditory stimuli were the sentence such as: *V kafe sideli pjat' X_{GEN.PL.}* ('In the café five X were sitting')². In this sentence, obstruents are located in a *non-neutralizing* position. A sample sentence as an auditory stimulus is given in (1). A preliminary study confirmed that the voicing contrast in the auditory stimuli was very easily perceived for the native listeners of Russian (10 listeners, identification task, average d' : 4.135, SD: 0.2)³.

Next, in response to each auditory stimulus, the participants were asked to produce the target word with the frame sentence such as: *Odin X_{NOM.SG.} el pečen'e.* ('One X_{NOM.SG.} was eating a cookie.'). In the frame sentence, the participants were expected to generate a nominative singular form, where the target obstruent is located in a neutralizing position, i.e., phrase-final position. A sample sentence is shown in (2).

(1) *V kafe sideli pjat' pri[g]-ov* ('In the café five *prig-GEN.PL.* were sitting.')

(2) *Odin pri[k] el pečen'e.* ('One *prig-NOM.SG.* was eating a cookie.')

The 60 target items and 60 fillers were presented in a pseudo-random order with four blocks: minimal pairs never appeared within the same block or the flanking blocks. Items were randomized within each block. Stimuli were presented with a computer program created by Praat (Boersma and Weenink 2010). Before the experiment, the participants worked through three demonstrations with existing words and seven practice runs with pseudo words.

² For the sake of simplicity, a simple romanized transcription of Cyrillic alphabet is used here, rather than phonetic transcription.

³ Those listeners did not participate in the production experiment.

The participants were encouraged to imagine that they would continue a story according to the Russian sentence (i.e., the auditory stimulus) they heard.

The recordings were conducted in a quiet room in Orenburg State University and the university dormitory. The tokens were recorded onto a portable recorder (SONY PCM-M10) with a stereo microphone (SONY ECM-MS907) using a 44.1 kHz sampling rate at a 16 bit quantization level. The first channel of the stereo recordings was extracted for the acoustic analysis.

2.5 The data set

In total, 720 tokens were elicited from the speakers (60 tokens $\times$ 12 speakers). Out of 720, 30 tokens (4.1%) were discarded because of pronunciation errors or recording errors. The remaining 690 tokens were submitted for the acoustic analyses.

2.6 Acoustic analysis

The target words were acoustically analysed using the Praat speech analysis software (Boersma and Weenink 2010). The following temporal and frequency properties, which are associated with a voicing contrast (e.g., Kingston and Diehl 1994), were measured:

VI: Duration of preceding vowel;

Voicing: Voicing duration during closure/frication;

C: Constriction (closure or frication) duration;

R: Release duration (of stop consonants);

F0 pre C: Fundamental frequency (F0) at the edge of the preceding vowel;

F1 pre C: First formant (F1) at the edge of the preceding vowel.

The representative waveforms and spectrograms are given in Figure 1 to illustrate the measurement protocol. For *VI*, when a nasal, lateral or fricative preceded the vowel, the onset of the vowel was defined as the time point of abrupt change in amplitude of the waveform. Where a trill preceded, the vowel onset was defined as the end of a short silence. The offset of the vowel was identified from an abrupt change in amplitude and the shape of the waveform.

Voicing duration during closure/frication was defined as an interval between the offset of the preceding vowel and the end of periodicity in the waveform.

For *C*, Closure duration was measured from the vowel offset to the onset of the burst noise. Frication duration was measured from the vowel offset until the end of fricative noise with an abrupt change in the amplitude and the spectrogram.

For *Release*, the release portion duration of stops was measured from the onset of the burst until the end of noise or an abrupt change in the waveform and spectrogram.

For *F1 and F0 pre C*, First formant (F1) at the edge of the preceding vowel was extracted by using the Burg algorithm of Praat's program. Fundamental frequency (F0) was extracted by using the autocorrelation algorithm. The tokens in which the F0 was not correctly extracted due to irregular vocal fold vibration were excluded from the F0 analysis.

Some tokens were excluded due to the difficulty of defining the segmental boundary. Two pairs (/ses/- /sez/, /dnek/-/dneg/) were systematically excluded from the F1 analyses since some participants produced the onset consonant with palatalization, while others produced it as an unpalatalized one.

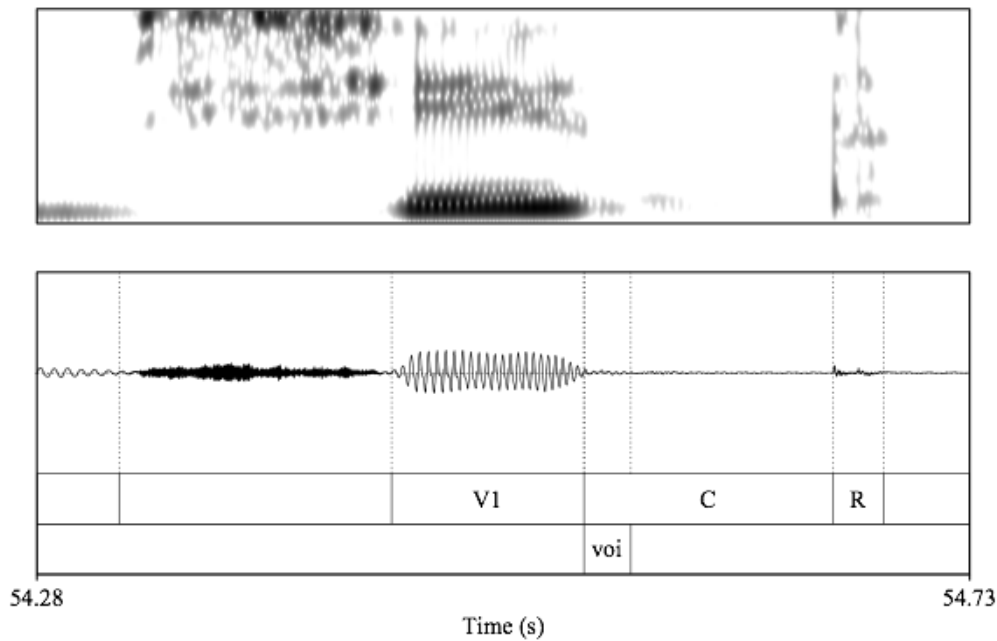


Figure 1
A representative spectrogram and waveform (/zid/).

2.7 Statistical analysis

Linear mixed-effect model was performed for each measurement. Underlying Voicing (Uvoi) and Place of Articulation (POA) served as fixed factors, and participants and items as random factors, assuming a random intercept⁴. In each model, p values are estimated by likelihood-ratio tests. Since separate acoustic dimensions were assessed at the same time, significance level α was adjusted for multiple comparisons by using Bonferroni correction. Therefore, instead of using the significance level $\alpha = 0.05$, the following levels of significance were used: $\alpha_{\text{adjusted}} = 0.008$ ($0.05/6$) for stops, $\alpha_{\text{adjusted}} = 0.01$ ($0.05/5$) for fricatives.

⁴ We compared the models assuming random intercepts only with those assuming both random intercepts and random slopes. Since no statistically significant differences were found between those models for all measurements, we use random-intercept-only models throughout the statistical analysis.

2.8 Predictions

If the neutralization is incomplete, it is predicted that there are some differences between “devoiced” and voiceless obstruents. The acoustic analysis confirmed that the speaker who provided the auditory stimuli signalled durational and spectral patterns in the stimuli, in which the target obstruents are in a non-neutralizing (intervocalic) position (Table 1).

Table 1
Summary of observed pattern in a *non*-neutralizing (intervocalic) position.

Stop			Fricative		
Measurement	Voiceless	Voiced	Measurement	Voiceless	Voiced
<i>VI</i> (ms)	114(19)	126(16)	<i>VI</i>	122(17)	150(26)
<i>Voicing</i> (ms)	14(5)	63(10)	<i>Voicing</i>	13(5)	63(13)
<i>C</i> (ms)	90(12)	63(10)	<i>C</i>	133(16)	65(14)
<i>Release</i> (ms)	18(6)	8(4)	<i>F0 pre C</i>	195(6)	190(7)
<i>F0 pre C</i> (Hz)	194(5)	192(9)	<i>F1 pre C</i>	484(157)	432(69)
<i>F1 pre C</i> (Hz)	465(121)	460(95)			

The directions of these patterns are consistent with those cross-linguistically reported (Kingston and Diehl 1994, among others). Our prediction is that at least one of the above-mentioned acoustic patterns is found in the neutralizing environment. However, if none of these acoustic dimensions show a difference, we may conclude that neutralization is complete (in this environment in Russian).

3. Results

3.1. Stops

The word-final stops in the dataset were typically realized with short vocal fold vibration during stop closure (Voicing proportion during stop closure: Mean = 0.16, SD = 0.12), suggesting that the underlying voiced tokens underwent final devoicing. The representative tokens are illustrated in figure 2a and 2b.

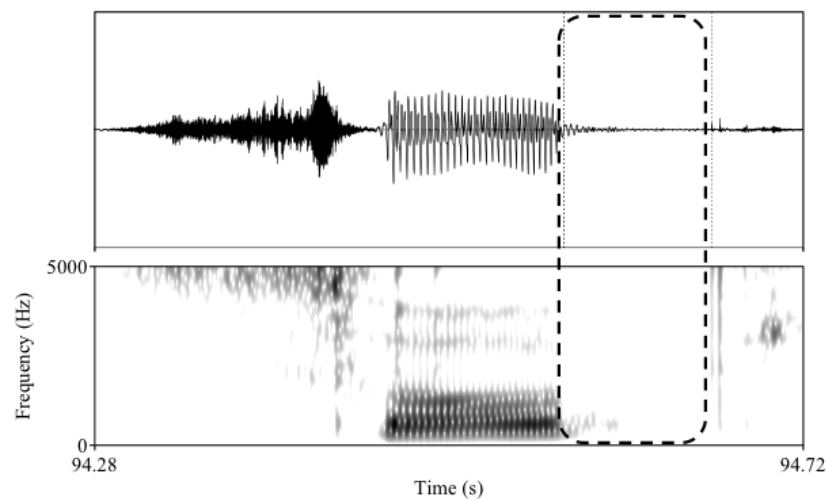


Figure 2a

A representative token of word-final voiceless stops (/tsot/). Closure portion is indicated by a broken line.

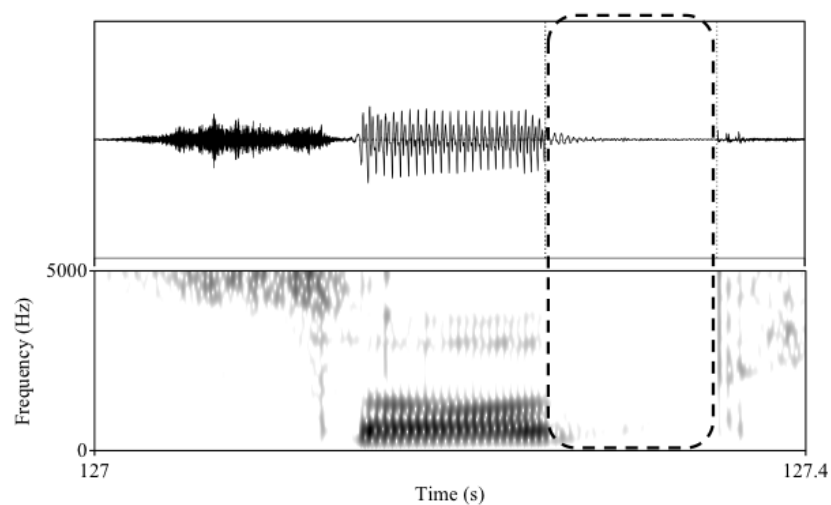


Figure 2b

A representative token of word-final devoiced stops (/tsod/). Closure portion is indicated by a broken line.

Table 2

Average values with one standard deviation (in parentheses) for measurements for stops. The measurement with statistical significance is indicated in boldface.

	Bilabial		Dental		Velar	
	Voiceless /p/	Voiced /b/	Voiceless /t/	Voiced /d/	Voiceless /k/	Voiced /g/
<i>V1</i> (ms)	106(18)	112(17)	106(25)	111(21)	106(19)	115(23)
<i>Voicing</i> (ms)	13(5)	16(7)	15(11)	17(13)	14(8)	16(9)
<i>C</i> (ms)	115(20)	114(22)	93(20)	95(23)	92(20)	92(18)
<i>Release</i> (ms)	54(29)	47(25)	52(21)	48(24)	61(25)	64(31)
<i>F0 pre C</i> (Hz)	239(72)	238(76)	238(70)	236(69)	241(74)	244(75)
<i>F1 pre C</i> (Hz)	453(108)	454(104)	450(106)	445(108)	428(123)	422(104)

Table 2 summarizes the average values for each phonetic dimension that is measured. The linear mixed-effect model indicated a significant main effect of underlying voicing in the duration of the preceding vowel [estimated difference: 5.1 ms, $X^2(3)=19.207$, $p<0.001$]. The vowels preceding underlying voiced stops were longer than those preceding underlying voiceless stops. There was no significant main effect of Place of Articulation (POA) on *V1* [$X^2(4)=1.9184$, $p=0.7508$, n.s.] nor interaction [$X^2(2)=1.8199$, $p=0.4025$, n.s.]. For the other phonetic dimensions (*Voicing*, *C*, *Release*, *F0 pre C* and *F1 pre C*), the main effect of underlying voicing contrast did not reach significance.

As for the difference in POA, the main effect of POA was significant for closure duration [$X^2(4)=27.529$, $p<0.001$] and release duration [$X^2(4)=16.687$, $p<0.002$]. Post-hoc analyses indicated that the bilabial stop had significantly longer closure duration than dental

[estimated difference: 20.5 ms, Tukey test, $p < 0.001$] and velar stops [estimated difference: 22.3 ms, Tukey test, $p < 0.001$]. The velar stop had significantly longer release duration than dental [estimated difference: 11.5 ms, Tukey test, $p < 0.001$] and bilabial stops [estimated difference: 12.6 ms, Tukey test, $p < 0.001$]. The results of the statistical analysis are summarized in Table 3.

Table 3

Summary of the statistical analysis. P values smaller than 0.01 (after being adjusted for multiple comparisons, by Bonferroni correction) are indicated as ** ($p < 0.008$) or *** ($p < 0.001$).

Dependent variable	Fixed effect Underlying voicing	Interaction Uvoi: POA
<i>VI</i>	***	n.s.
<i>Voicing</i>	n.s.	n.s.
<i>C</i>	n.s.	n.s.
<i>Release</i>	n.s.	n.s.
<i>F0 pre C</i>	n.s.	n.s.
<i>F1 pre C</i>	n.s.	n.s.
	Fixed effect POA	Multiple comparison POA
<i>VI</i>	n.s.	
<i>Voicing</i>	n.s.	
<i>C</i>	***	Bilabial vs. Dental*** Bilabial vs. Velar*** Velar vs. Dental, n.s.
<i>Release</i>	**	Velar vs. Bilabial*** Velar vs. Dental*** Bilabial vs. Dental, n.s.
<i>F0 pre C</i>	n.s.	
<i>F1 pre C</i>	n.s.	

3.2. Fricatives

Similar to the case of stops (Section 3.1), the word-final fricatives in the dataset were typically realized with short vocal fold vibration during frication portion (Voicing proportion

during frication: Mean = 0.10 ms, SD = 0.06), suggesting that the underlying voiced fricatives underwent final devoicing. Representative tokens are illustrated in figures 3a and 3b.

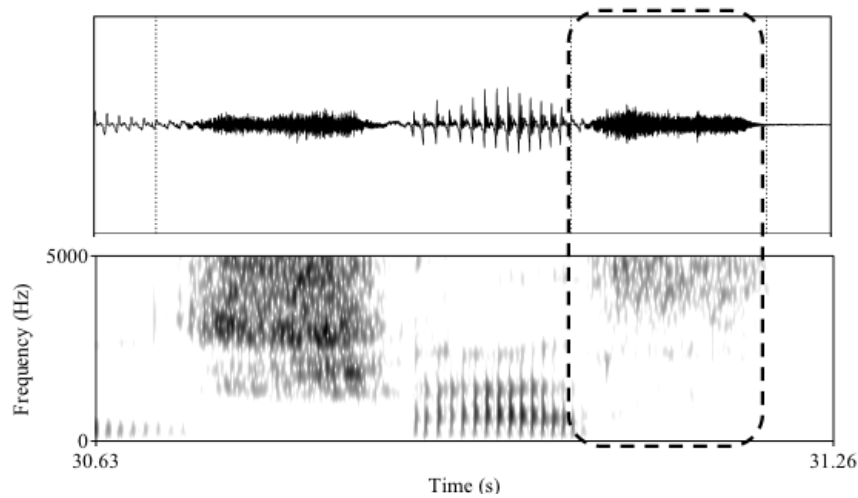


Figure 3a
A representative token of voiceless fricatives (/fas/).
Frication portion is indicated by a broken line.

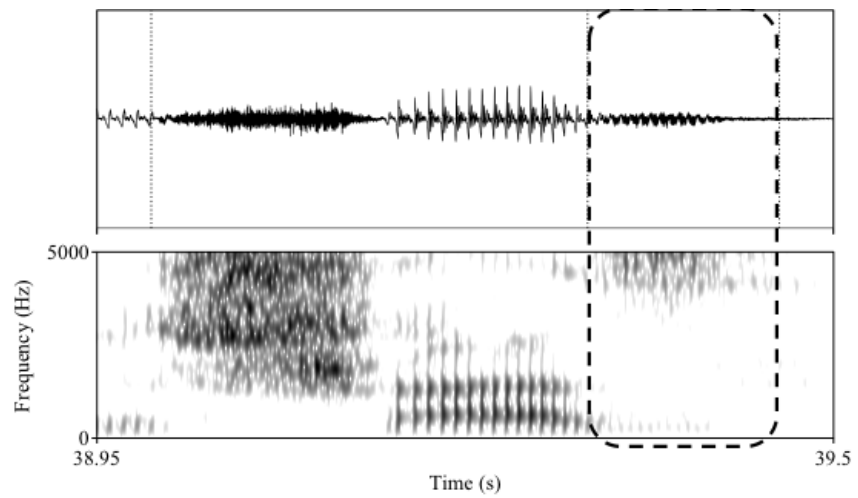


Figure 3b
A representative token of “devoiced” fricatives (/faz/).
Frication portion is indicated by a broken line.

Table 4 summarizes the average values for each measurement for quantitative analysis.

Table 4

Average values with standard deviations (in parentheses) for all measurements for fricatives. The measurement with statistical significance is indicated in boldface.

	Labiodental		Alveolar		Post alveolar	
	Voiceless /f/	Voiced /v/	Voiceless /s/	Voiced /z/	Voiceless /ʃ/	Voiced /ʒ/
<i>VI</i> (ms)	122(22)	125(21)	123(21)	126(26)	124(20)	130(23)
<i>Voicing</i> (ms)	15(7)	17(8)	11(6)	16(13)	14(5)	16(7)
<i>C</i> (ms)	150(30)	141(34)	163(29)	155(30)	165(29)	163(26)
<i>F0 pre C</i> (Hz)	241(72)	243(75)	247(77)	248(79)	250(74)	248(73)
<i>F1 pre C</i> (Hz)	455(130)	459(119)	422(110)	423(98)	453(132)	445(119)

A significant main effect of underlying voicing was found for the duration of voicing during frication [$X^2(3)=15.89$, $p<0.01$]. The voicing duration during frication was longer during underlying voiced fricative than during underlying voiceless fricative [estimated difference: 4.8 ms]. The main effect of POA on voicing duration was not significant [$X^2(4)=7.2546$, $p=0.123$], nor was their interaction [$X^2(2)=2.9364$, $p=0.2303$]. For any other measurements (*VI*, *C*, *F0 pre C*, *F1 pre C*), the main effect of underlying voicing was not significant.

The main effect of POA was significant for frication duration [$X^2(4)=14.767$, $p<0.01$]. Post-hoc analyses indicated that the labiodental fricative was significantly shorter in frication duration than alveolar [estimated difference: 13.6 ms, Tukey test, $p<0.01$] and post-alveolar fricatives [estimated difference: 18.5 ms, Tukey test, $p<0.001$]. Table 5 shows the summary of the statistical analysis for fricatives.

Table 5

Summary of the statistical analysis for fricatives. P values smaller than 0.01 (after being adjusted for multiple comparisons, by Bonferroni correction) are indicated as **($p < 0.01$) or *** ($p < 0.001$).

<i>Dependent variable</i>	Fixed effect Uvoi	Interaction Uvoi: POA
<i>Vl</i>	n.s.	n.s.
<i>Voicing</i>	**	n.s.
<i>C</i>	n.s.	n.s.
<i>F0 pre C</i>	n.s.	n.s.
<i>F1 pre C</i>	n.s.	n.s.
	Fixed effect POA	Multiple comparison POA
<i>Vl</i>	n.s.	
<i>Voicing</i>	n.s.	
<i>C</i>	**	Labiodental vs. Alveolar** Labiodental vs. Postalveolar*** Alveolar vs. Postalveolar, n.s.
<i>F0 pre C</i>	n.s.	
<i>F1 pre C</i>	n.s.	

4. Discussions

The aim of the present study was to document how word-final devoicing is implemented in utterance-medial position in Russian.

The results showed that there were significant acoustic differences between the words containing underlyingly voiced and those containing underlyingly voiceless obstruents. For stops, the duration of the vowel preceding the underlyingly voiced stop was significantly longer than that preceding the underlyingly voiceless stop. For fricatives, voicing duration during frication was significantly longer in the underlyingly voiced fricative than that in the underlyingly voiceless fricative. The directions of the difference observed in the neutralizing position were the same as those in the non-neutralizing position (See Table 1). The results of the present experimental task in general confirm that incomplete neutralization is also

observed in utterance-medial word-final position, in which speakers inflected pseudo nouns in response to the auditory stimuli. Incomplete neutralization exists even when orthography or lexical frequency effects are factored out.

Our results are in line with those of the previous studies in Russian, which found incomplete neutralization in existing Russian words (Chen 1970, Dmitrieva et al. 2010, Kharlamov 2012, Kulikov 2012). The magnitude of the difference observed in the present study is similar to that in Kulikov (2012), who examined final devoicing in stop consonants in an utterance-medial position. The difference in preceding vowel duration for stops is 6 ms in Kulikov (2012) and 5.1 ms in the present study. However, while Kulikov (2012) observed a difference in F1 at the edge of the preceding vowel, no differences were found in the present study. This may be due to the difference in the experimental tasks: Kulikov examined existing words by using a reading task, while this study exploited the pronunciation of pseudo words in response to auditory stimuli. This in turn suggests that in addition to underlying voicing, other factors may contribute to how incomplete neutralization actually manifests itself in the condition involving existing words, such as orthography and linguistic knowledge of lexical items. We hasten to add that this is a post-hoc specification which needs to be further examined.

To put the current findings into a broader perspective, Table 6 shows a brief survey of the acoustic studies of word-final devoicing in Russian. As can be seen in Table 6, Kulikov (2012) and the present study show similar results; that is, they both found the effect of underlying voicing on the preceding vowel duration. On the other hand, Dmitrieva et al. (2010) and Kharlamov (2012) found no significant difference both in preceding vowel. Instead, they found the difference in closure/frication duration and release duration. This difference may come from the difference in prosodic conditions examined: while Dmitrieva et

al. (2010) and Kharlamov (2012) asked the speakers to produce the words in isolation (i.e., utterance-final position), Kulikov (2012) and the present study embedded the words in the

Table 6

Comparison with previous acoustic studies in Russian

Study	Prosodic condition	Obstruent	Task	Acoustic cues to underlying voicing	Notes
Chen (1970)	Utterance -final	/p, b, t, d, k, g/	Orthographic	Preceding vowel duration	1 speaker, No statistical analysis
Pye (1989)	Utterance -final	/p, b, t, d, k, g, f, v, s, z, ʃ, ʒ/	Orthographic	Preceding vowel duration, voicing duration, Constriction duration, Release duration	No statistical analysis
Shrager (2010)	Utterance -final	/t, d/	Reading	Constriction duration, Release duration	
Dmitrieva et al. (2010)	Utterance -final	/p, b, t, d, k, g, f, v, s, z, ʃ, ʒ/	Orthographic	Constriction duration, Release duration	Both for monolingual and bilingual speakers. Token word frequency was controlled (real words)
Kulikov (2012)	Utterance -medial	/p, b, t, d, k, g/	Orthographic	Preceding vowel duration, F1	
Kharlamov (2012)	Utterance -final	/p, b, t, d, k, g, f, v, s, z, ʃ, ʒ/	Orthographic, Non-orthographic	Constriction duration, Release duration	Token word frequency was controlled (real words)
Current study	Utterance -medial	/p, b, t, d, k, g, f, v, s, z, ʃ, ʒ/	Non-reading	Preceding vowel duration; Voicing duration during frication	Token word frequency was considered (pseudo nouns)

middle of a sentence (i.e., utterance-medial position). These observations suggest that the type of the prosodic boundary after the word-final obstruent may affect the acoustic profile of word-final devoicing. The effects of prosodic boundary on phonetic details have been receiving attention in the recent literature (e.g., Kuzla and Ernestus 2011). Future studies could further address the effects of prosodic conditioning on word-final devoicing in Russian.

Finally, modelling incomplete neutralization in phonological theory is another issue for future consideration. Incomplete neutralization has been modelled both within the generative framework (e.g., Braver 2013) and within exemplar-based frameworks (e.g., Ernestus and Baayen 2007, Kirby 2010). More empirical findings could contribute to the discussion of which framework is better suited to model incomplete neutralization.

4. Conclusions

The present study provided additional data regarding the acoustic manifestation of word-final devoicing in Russian. The results showed that there were small but significant durational differences between words with underlying voiced obstruents and those with underlying voiceless obstruents, showing the case of incomplete neutralization.

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To be added.

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Appendix

Materials for the production experiment (a phonological transcription).

Speech materials for stops (pseudo nouns by definition)

Bilabial		Dental		Velar	
Voiceless	Voiced	Voiceless	Voiced	Voiceless	Voiced
rip	rib	zit	zid	prik	prig
lep	leb	ʒet	ʒed	dnek	dneg
tʃap	tʃab	xrat	xrad	tsak	tsag
nop	nob	tsot	tsod	zok	zog
mup	mub	ʃtut	ʃtud	bruk	brug

Speech materials for fricatives (pseudo nouns by definition)

Labiodental		Alveolar		Post alveolar	
Voiceless	Voiced	Voiceless	Voiced	Voiceless	Voiced
krif	kriv	zis	ziz	priʃ	priʒ
tsef	tsev	ses	sez	neʃ	neʒ
naf	nav	ʃas	ʃaz	plaʃ	plaʒ
ʒof	ʒov	dros	droz	moʃ	moʒ
ruf	ruv	plus	pluz	luʃ	luʒ